



Online Book Store Management System using MERN Stack

SmartBridge Internship Final Project

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Program

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Date

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Introduction & Problem Statement



Why online bookstores matter

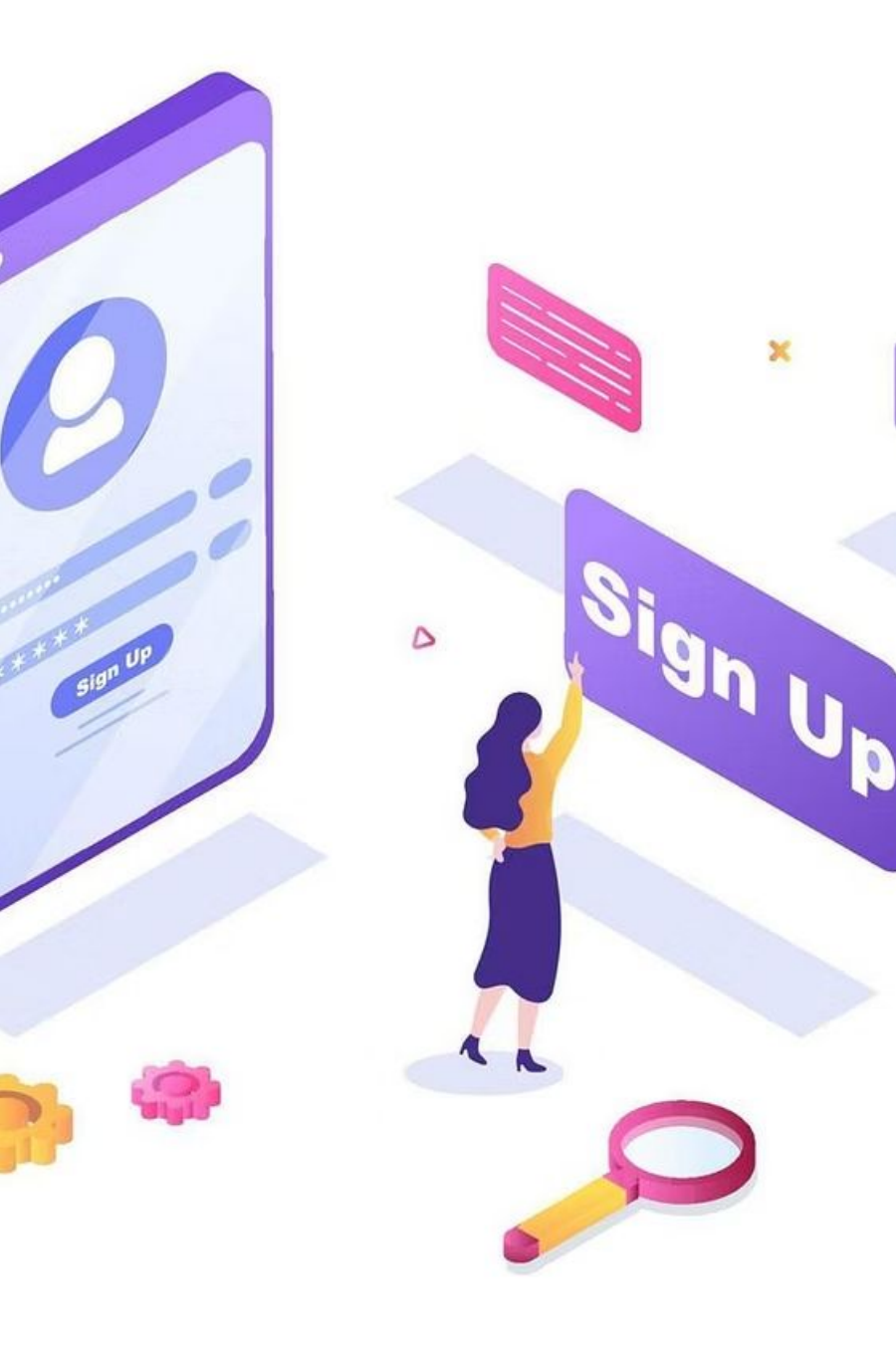
Provide 24/7 access to catalogs, wider selection, and convenient delivery — expanding reach beyond physical stores.



Problems with traditional stores

- Manual inventory management
- Limited accessibility and hours
- Time-consuming search for books
- Inefficient order handling

Need: A digital, secure, and responsive platform to streamline search, inventory, orders, and admin workflows.



Project Objectives

- Simple user registration and secure login (JWT authentication)
- Fast browse & search with category filters
- Shopping cart with add/remove and secure checkout
- Admin controls for books, inventory, and orders
- Responsive, user-friendly interface across devices

Technology Stack



MongoDB

Document database for flexible schemas and fast queries



Express.js

Lightweight API framework for routing and middleware



React.js

Component-based frontend for responsive UI



Node.js

Server runtime for API and real-time features



JWT

Stateless, secure authentication tokens



HTML5 / CSS3 / JS

Modern web standards and client scripting



Git & GitHub

Version control and collaboration

System Architecture

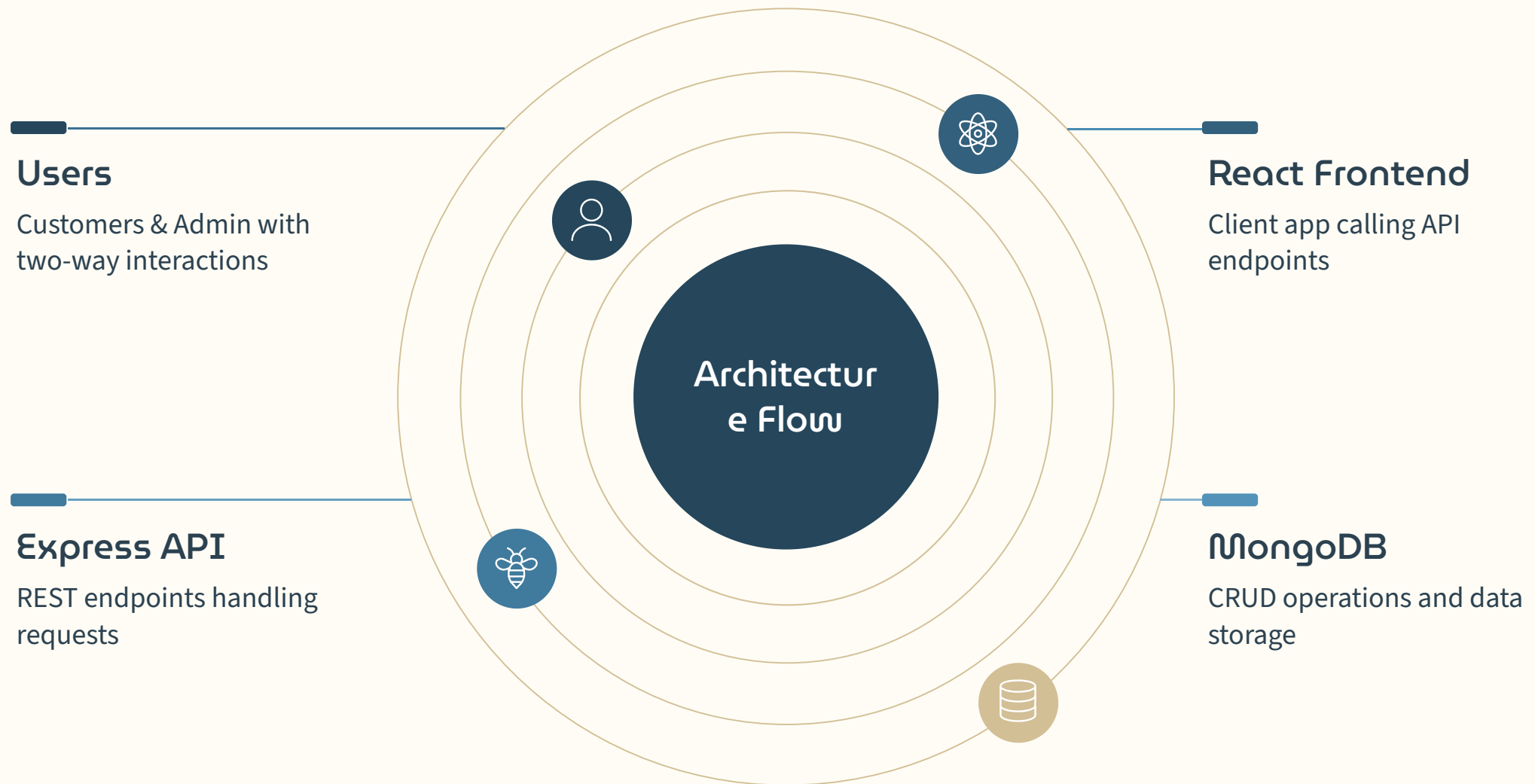


Diagram shows clear flow: User → React Frontend → Express API → Node.js Server → MongoDB. Admin and Customer both interact with backend endpoints; admin has protected routes for management.

Modules & Workflow

User Authentication

Registration, login, JWT, role-based access

Book Management

CRUD books, categories, stock updates

Search & Categories

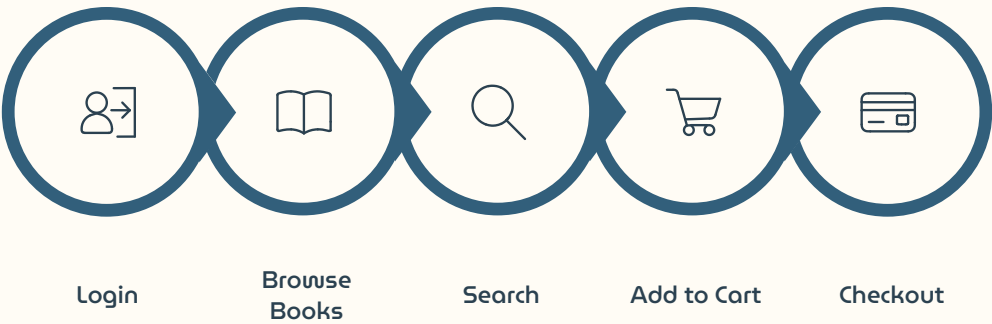
Full-text search, filters, sorting

Shopping Cart & Orders

Session/cart persistence, order workflows

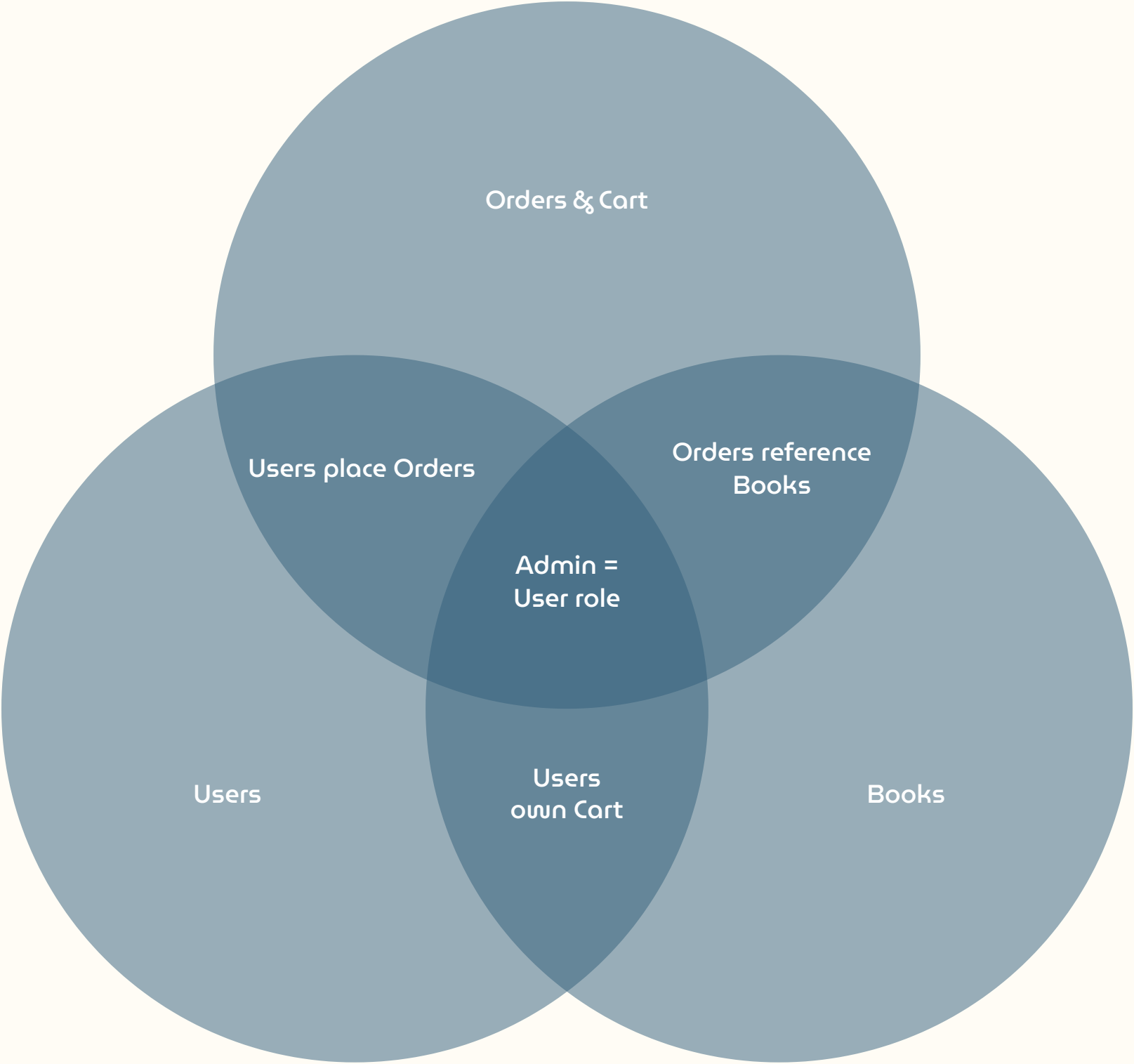
Admin Dashboard

Order processing, inventory and user management



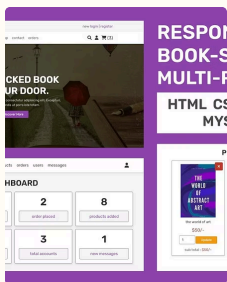
Workflow maps typical customer journey and highlights integration points for inventory checks and order creation on checkout.

Database Design (ER Diagram)



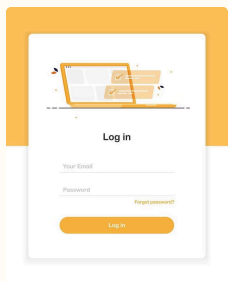
Relational structure uses references: Orders and Cart store book references and quantities. Users store role flags to differentiate admin and customer.

Key Features & UI Screens



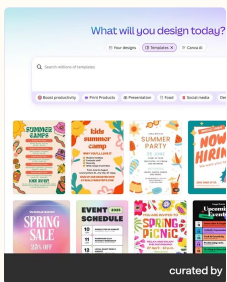
Home Page

Featured books, search bar, category sections



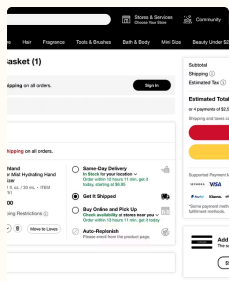
Login / Registration

Secure JWT login with validation



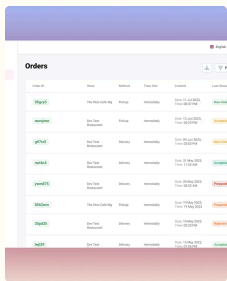
Book Listing & Filters

Category filter, sorting, and search



Cart & Checkout

Modify quantities, secure checkout flow



Admin Dashboard

Manage books, process orders, view reports

All screenshots are placeholders for editable UI mockups to be included in the final deliverable.

Results & Future Enhancements

- **Results:** Streamlined online purchasing and catalog browsing
- Efficient inventory and order management for admins
- Faster order processing and improved UX across devices
- Secure authentication and role-based access
- **Future Enhancements:**
- Integrate online payment gateway (PCI compliant)
- AI-driven book recommendations and personalised feeds
- Wishlist, reviews & ratings, and email notifications
- Native mobile app for iOS & Android



- ❏ Planned roadmap prioritises secure payments and recommendation engine for increased conversion and retention.



Conclusion & Thank You

The MERN-based Online Book Store Management System delivers a secure, scalable, and user-friendly platform that simplifies book management and enhances the shopping experience for customers and administrators.

Questions & Answers

Please share feedback or questions
— eager to discuss implementation
details and next steps.

Contact

Student Name — Email / GitHub
repository (include in final handoff)